Navigating Alberta's Electricity Market

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BTMA 531: Data Analytics Tools for Business
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April 8, 2026

Problem Formulation

Alberta operates a unique electricity market that is different from the rest of Canada, functioning as an "energy only" market. The Alberta Electric System Operator (AESO) is responsible for the safe and reliable operation of the Alberta Interconnected Electric System (AIES). In this report, predictive analysis will be performed on multiple datasets to investigate AESO's pool price and metered volume data for wind and solar assets. This will help determine how profitable solar and wind assets are, and if they are a worthy investment. This analysis will also help large energy consumers and citizens understand how energy usage at different points of the day are priced differently.

The AESO pool price is the hourly wholesale clearing price for electricity, representing the average of 60 one-minute System Marginal Prices (SMP). It is important for generators to track the pool price because it is the market price in real time that they are being paid for to supply electricity. Therefore, this represents the electricity demand in real time. Investigating patterns of the pool price can help one detect energy highs and lows within a day, week, or year. This can be useful to businesses who consume high amounts of energy, or to your average citizen using electricity at home. Therefore, having an understanding can benefit large energy consumers and households to make more informed energy usage decisions.

In addition, the AESO is also responsible for collecting hourly metered volume data, which measures the quantity of electric energy (in MWh) from generators. This is another crucial aspect to understanding the electricity market because it is documenting how much energy is being used, by which generators, and for how long. The pool price overlaps with this because generators are paid the pool price for every MWh they produce (see Appendix). Renewable energy is up and coming in today's society as coal becomes less common. Therefore, it is important to track how well these renewable assets, like wind and solar, are performing. Thus, tracking the real time energy generators produce can be helpful for businesses looking to invest in renewable energy or citizens who want to buy a solar panel and sell energy back to the grid. Hence, this data can help estimate how well one's investment in renewables would perform.

Pool Price Data

Alberta Electric System operator retains historical pool price data along with Alberta internal load. We extracted this dataset with 54 thousand observations across January 2020 to 2026 February into multiple csv files. Our goal was to train, test, predict pool prices and forecast future values for the next hour, next day, and next week through machine learning methods. We believe such predictions will have significance in forecasting Alberta pool prices and understanding patterns for traders and pool participants. The data was then organized and simplified through verifying variable types. This included date, pool price, 30-day rolling average and internal load (AIL). Using RStudio, the multiple CSV files were combined by using the function *cbind()*. To further organize the data, the Datetime column hours were put into a descending order and plotted against pool price to preview existing patterns. After the data had been cleaned, it was exported into a CSV file for the model.

Metered Volumes Data

Metered Volume data was collected from the AESO website by going to the "Historic" section and downloading the "Metered Volumes (All)" dataset for 2025. Collecting specific wind and solar assets was done through the reporting portal (ets.aeso.ca) and manually transferred over to a table in excel that contained the energy type, asset ID, and MC values. This was later used to filter the metered volumes dataset. During the data cleaning process, the date information was reorganized into a dedicated column in Excel. The dataset and wind/solar asset table was then imported into PowerBI, where empty rows and columns were removed. Using query relationships, the data was then filtered to only include wind and solar assets. The cleaned and filtered data was then exported to a CSV file used for modeling in RStudio.

Feature Analysis on Pool Price & Wind/Solar Assets

To understand what drives Alberta pool prices, we looked at key variables using correlation, simple regression, and an XGBoost model.

Net Load shows a moderate positive relationship with price, with a correlation of 0.41 . This means that when demand is high relative to supply, prices tend to increase. The regression results support this, showing that a 1000 MW increase in Net Load leads to about a $55/MWh increase in price, although it explains only a portion of the variation ($R^2 \approx 0.17$).

Wind generation has a moderate negative relationship with price (correlation = -0.34) . This suggests that more wind supply helps lower prices, with a 1000 MW increase in wind associated with roughly a $58/MWh decrease in price, though its overall impact is still smaller than demand.

There are also clear time-of-day patterns. Prices are lowest overnight (around $47–56/MWh) and increase significantly during peak hours, reaching as high as $130-160/MWh in the evening. This shows that electricity prices closely follow daily demand patterns.

The XGBoost model helps identify which features are most important for prediction. The most important variables are lagged prices (previous hour and previous day). This shows that electricity prices are highly dependent on recent past values. Among the non-lag variables, Net Load is the most important, confirming that demand is the main driver of price changes. Time-related features (like hour of day) also play a strong role. Wind generation has some influence in lowering prices, but it is not a dominant factor. Solar generation has very low importance, which matches the earlier analysis.

This model achieved an RMSE of 40.60. While its predictive accuracy is lower than our other model, its primary purpose was to identify the key factors influencing pool prices.

Pool Price Analysis

- **Transformation**

Following the cleaning of the dataset, we conducted some EDA, revealing that price distribution represented a right-skewed histogram (appendix A) with a mean of $90/MWh and a median of $42/MWh. Data had multiple zero values along with a max price of 999 in some cases. Using only raw prices, models failed to differentiate price drops or spikes, and provided outputs with recurring values under $1. Thus, we normalized the data by adjusting zero values and log-transforming the prices.

Once we plotted these shifted prices across the timeline (appendix B), proper signs of time-series data with frequent changes, spikes, and drops were revealed. But no signs of seasonality or recurring patterns, thus we decomposed the prices in weeks. The decomposed plot (appendix C) evidently reveals signs of seasonality, with a general trend of prices peaking due to the 2022-2023 energy shortage in Alberta.

Thus we could infer that datetime factors, hourly lags for Price and AIL, and spike detection are important features that should be injected into the model. Without these features, the models did not understand patterns in datetime (especially spikes and outliers). ACF/PACF confirmed the weekly and seasonality pattern (appendix D). Stationarity tests revealed data is highly stationary with a very low p-value. Thus, data did not require differencing.

- **Lags & Features**

(appendix E)

- **Models**

We created 4 models (Linear, ARIMA, ARIMAX, XGBoost) with more than 5 years of pool prices as training sets. Then we tested the last 30 days of data as the prediction horizon.

We started with the linear model and tested various lags and features to see which ones might be significant in results or training patterns. Thus, we refined the lags and features from the results of linear regression, then combined the best of them through multiple iterations.

Then we made the XGBoost model, as we intended from the start, with all the combined features since this method is suitable for handling so many variables. The linear model acted as the baseline for the selection of features, but we had to tune certain parameters manually to determine which combination had the lowest errors. Lastly, we used auto.arima() for the ARIMA and ARIMAX models. We opted out some of the lags from our combined features (derived from the feedbacks of the linear model) to achieve a better performance during ARIMAX.

However, we also had intentions to make a neural network, but that did not seem appropriate. And we attempted a basic Holt-Winters model too, as suggested by the professor. But the results were inconclusive and loaded with errors, therefore we decided to make the ARIMA & ARIMAX to compare against the traditional linear method and the machine learning XGBoost model.

- **Analysis**

We calculated RMSE and MAE of both test & train sets to record incremental improvements in residuals and to compare the models. We also included MASE to indicate comparison against the naive benchmark and an accuracy measure using mean of actual test prices. Below is the summary of the models' performances in terms of errors and accuracy:

Model	Train RMSE	Test RMSE	Train MAE	Test MAE	Accuracy
XGBoost	42.664	12.364	17.187	4.939	77.5%
Linear	47.895	19.682	24.113	10.411	52.5%
ARIMAX	76.003	38.169	29.439	15.314	30.1%
ARIMA	78.615	39.620	30.278	19.093	12.9%

Model performances were sorted using mean absolute error across training and test sets. The comparison table indicates that XGBoost achieved the best overall performance, with the lowest errors and highest predictive accuracy at 77.5%. Its Test MAE of 4.939 suggests predictions were, on average, within $5/MWh of actual prices. The low MASE (0.167) further demonstrates substantial improvement over the naive benchmark. However, accuracy decreased slightly when extending the prediction horizon to 2 or 4 months, short-term predictions (hourly to daily) remained highly reliable.

The Linear Regression model ranked second, achieving 52.5% accuracy, confirming that traditional approaches can still deliver competitive results, even when combined with multiple features and variables. Moreover, linear model insights were instrumental in refining lags and features, which in turn, informed the XGBoost & ARIMAX models through consistent iterations.

The ARIMA and ARIMAX models performed comparatively weaker, despite identifying seasonality and differencing the data, even after ADF suggesting that data needs no difference. Both models exhibited high residual errors and over-differencing tendencies. However, the inclusion of combined features (after some process of elimination) as exogenous regressors (xreg) slightly improved ARIMAX outcomes.

Plotted predictions (appendix F) revealed that the XGBoost model, with its test set values, closely followed actual price trends, with some underestimation of extreme spikes. It validates the low error margin and 77.5% accuracy in test predictions. Despite the low MAE, we plotted the residuals (appendix G) as well, revealing well-contained deviations capped near 150, consistent with the model's performance metrics relative to a maximum recorded price of 999. Overall, XGBoost demonstrated strong capacity to capture hourly price volatility and temporal dynamics, making it a reliable foundation for short-term AESO pool price predictions on existing data.

- **Forecasting**

Using the XGBoost model as the foundation, we extended predictions beyond the train/test period to generate short-term forecasts. Since future values for AIL and the 30-day rolling average were unavailable, a recursive loop was applied using lagged variables and placeholders for actual mean to simulate future inputs into the boosted model.

The model forecasted pool prices for the week of March 2–8, 2026, producing next-hour, next-day, and next-week estimates. We plotted the forecast (appendix H), showing strong alignment with recent actual prices initially but became smoother toward the forecast horizon, reflecting the influence of placeholder variables and the recursive loop.

These results highlight the model's capability to generate reasonable short-term forecasts. The smoothing effect towards the end of the plot indicates further scope for improvements. Due to time and resource constraints, we were unable to do a walk-forward validation or derive actual future variables for the forecasting. Thus, we retain ample scope to introduce backpropagation, readjust the parameters, iterate through error correction, and compare against actual pool prices of March 2026.

Challenges and Shortcomings

Several limitations arose when working with the data and modeling. For instance, there was a limitation of missing values when working with the metered volume data. Since not all generators operate continuously, the dataset contained gaps where generation values were absent for certain hours. Not every generator is running every hour and therefore looks missing according to the data sheet. To complete the data, generators that were not running were given a value of zero to show that they were not producing any energy. If further analysis were to be performed in the future, this would become a challenge in continuously updating data and assigning zero values.

Another limitation we came across is, while our initial XGBoost model helped identify the most significant drivers of pool prices, its predictive accuracy was reduced due to the inclusion of many variables, resulting in a higher RMSE of approximately 40.

In addition, in the forecasting component, the recursive forecasting approach introduced smoothing bias which reduces the model's ability to capture price spikes influencing the overall results. This limitation can be seen through the forecast plots, where the visual looks more uniform instead of capturing extreme behaviour, reducing the accuracy of the visual.

Results and Discussion

For the analysis performed on metered data, our first XGBoosting model (without time lag) showed that net load was the main driver of the pool price, while wind and solar production had a much smaller impact on the price. This suggests that renewable output alone does not strongly influence price levels in the current market structure, and that the main influence is energy demand. Therefore, our first model confirms that a higher demand leads to higher prices, especially during peak energy hours, such as the evening (7pm-10pm). For predictive modeling, the XGboost model significantly outperformed all the

other models. The XGBoost model overall achieved the lowest error and the highest accuracy. As well, XGBoost test predictions are less than $5 over or short of actual test prices.

From our findings, investing in wind and solar assets in Alberta can be viable, however, profitability depends more on timing and market conditions than on direct price influence. Since electricity prices are primarily driven by demand rather than renewable supply, businesses should focus on strategic participation during high-demand periods (peak hours) to maximize revenue. They should also combine renewable investments with price forecasting tools, such as the XGBoost model, to better decide when to sell electricity.

In addition, companies should consider using energy storage systems, which allow them to store excess energy and sell it later when prices are higher. For large energy consumers, the results suggest that shifting electricity usage away from peak hours can significantly reduce costs, since price spikes are closely tied to demand patterns rather than changes in renewable energy supply.

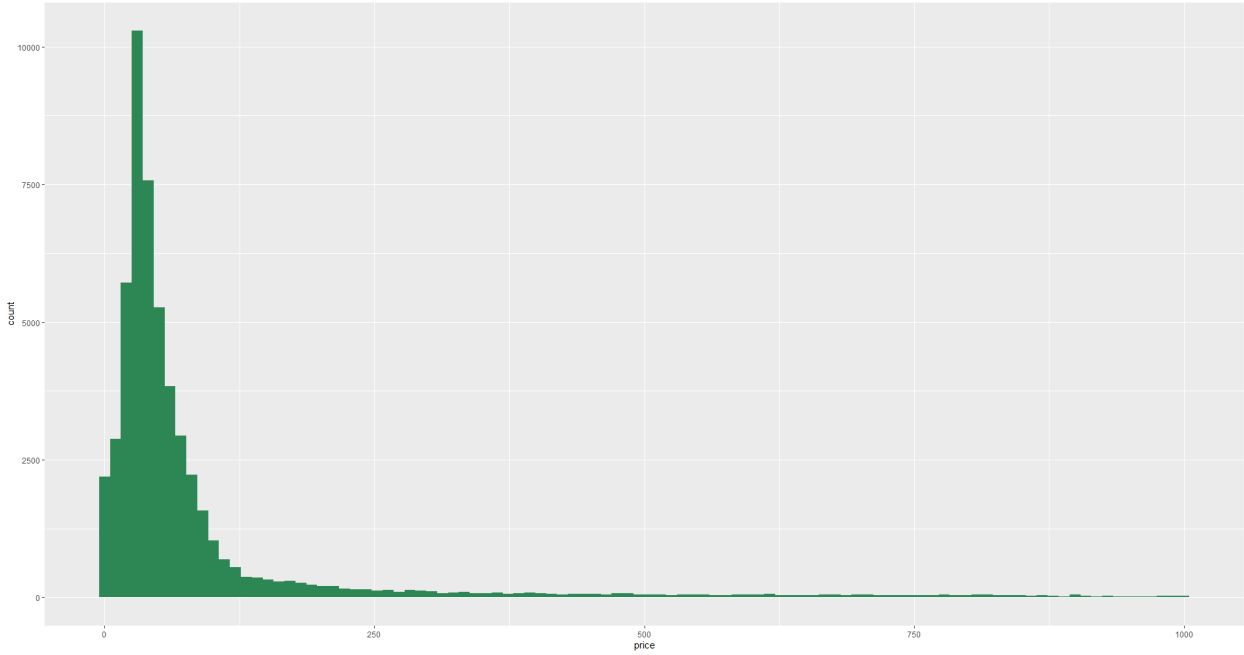
For future work, we suggest several ways to improve and extend this project. First, additional explanatory variables such as weather data, natural gas prices, and grid constraints should be included to provide a more complete analysis. Second, forecasting accuracy could be improved by predicting key input variables, such as AIL, alongside electricity prices. Third, implementing walk-forward validation would better reflect real-world forecasting conditions.

In addition, more advanced machine learning models, such as LSTM neural networks designed for time series data, could be explored. Improving data quality is also important, particularly by obtaining more complete and higher-resolution data on renewable energy generation.

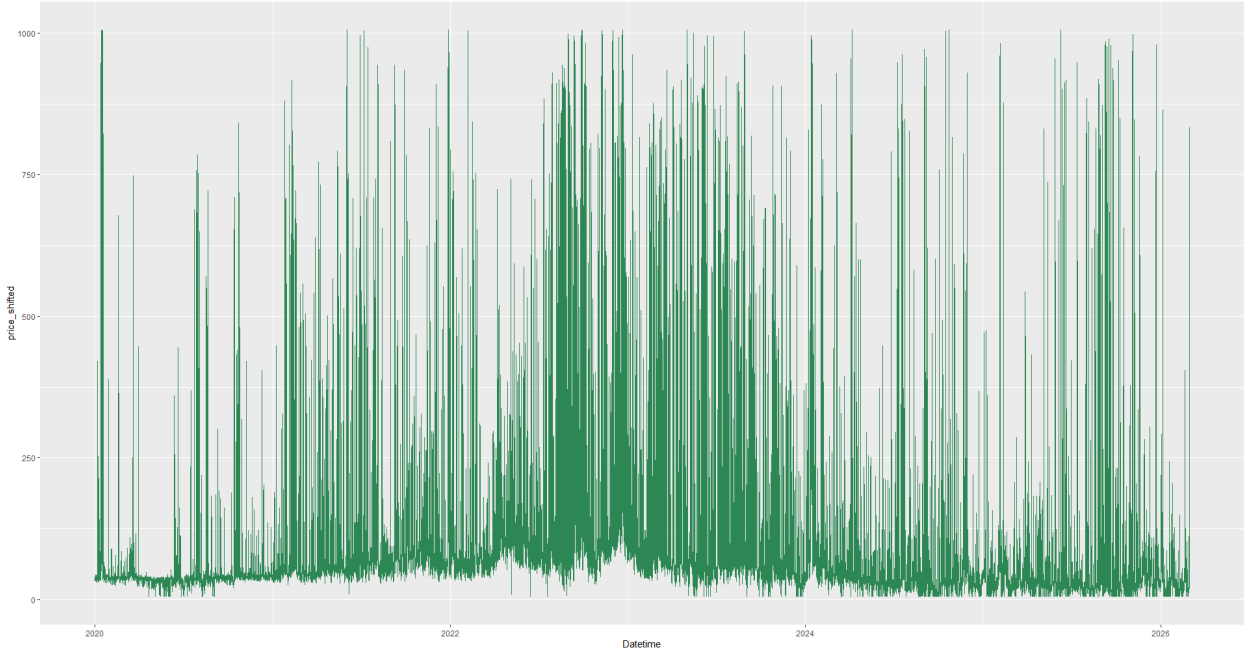
Overall, while renewable generation does not currently show a strong direct impact on pool prices, combining renewable investments with advanced predictive tools creates a valuable opportunity for both producers and consumers to make more informed and profitable decisions in Alberta's electricity market.

Appendices

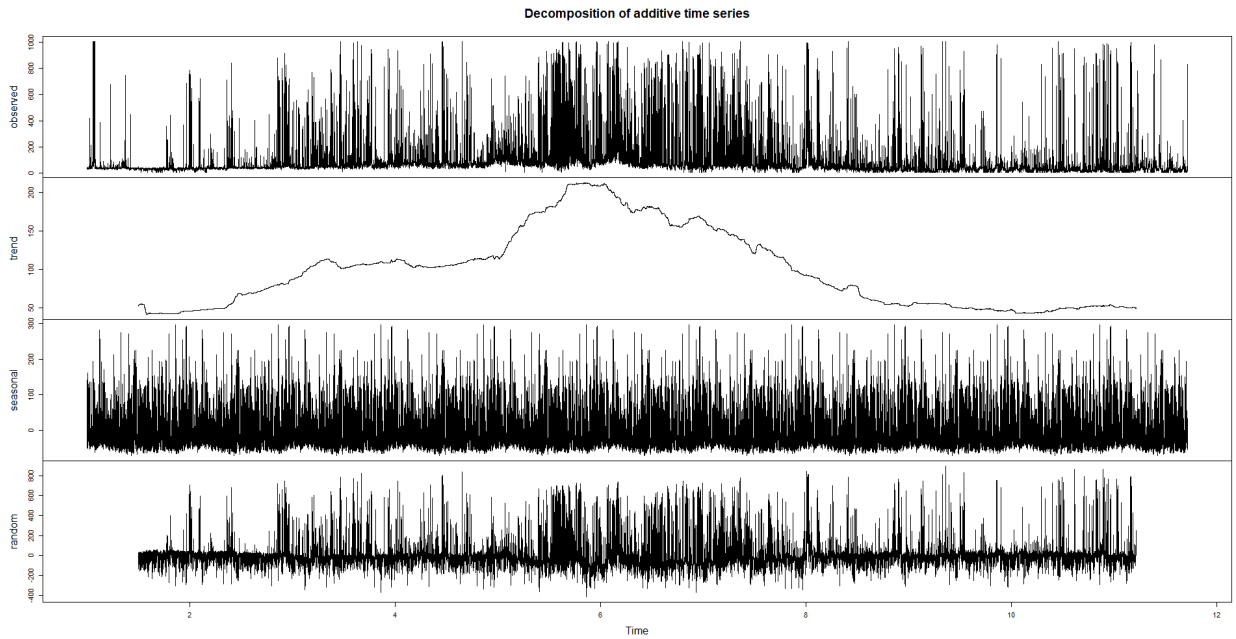
Appendix A: Raw price histogram



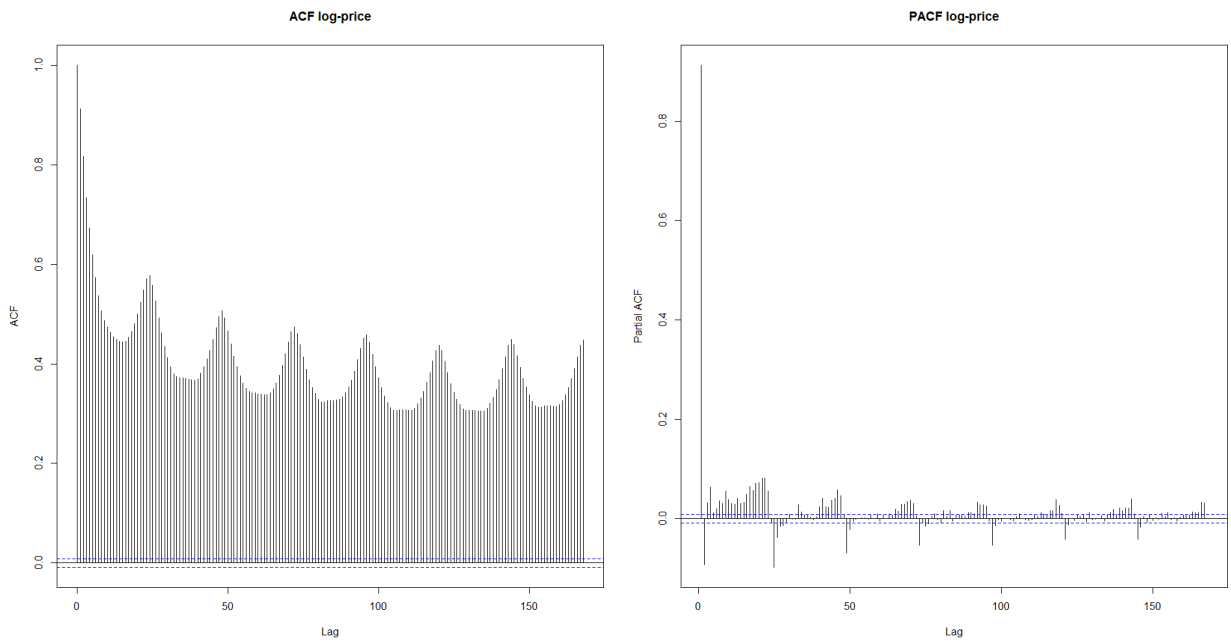
Appendix B: Raw time-series preview



Appendix C: Decomposed time-series for trends and seasonality



Appendix D: ACF & PACF test with log prices

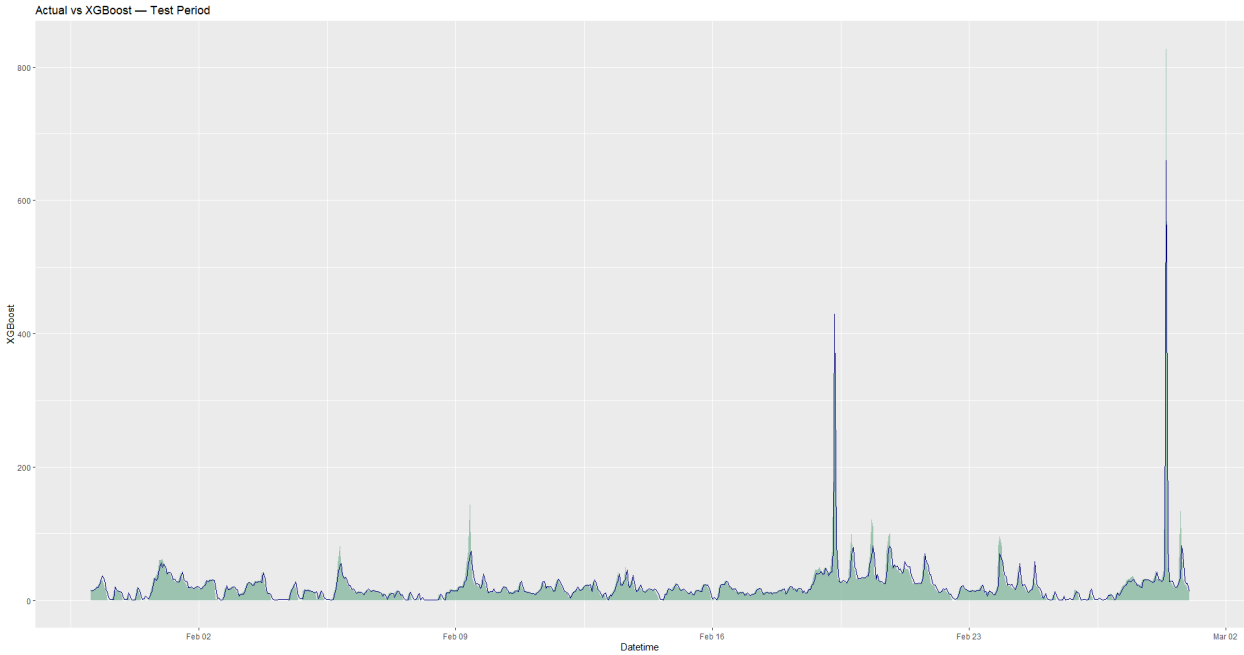


Appendix E: Features table consisting model inputs

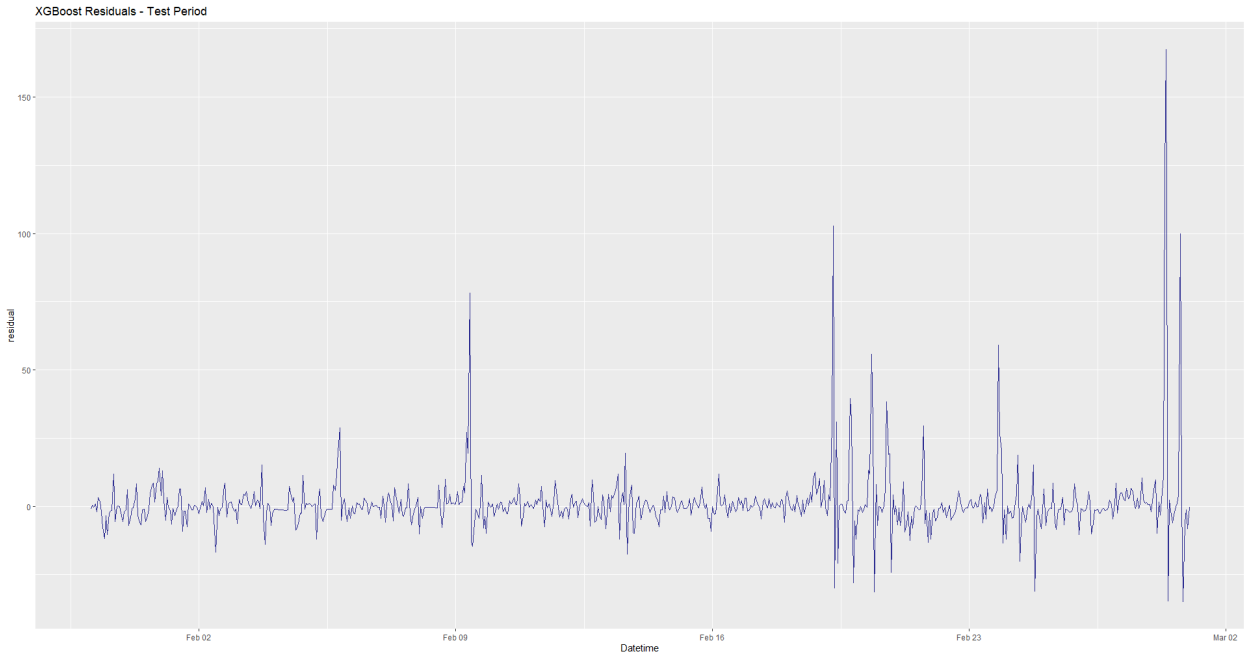
Features	***Type***	***Characteristics***	***Usage***
isweekend	binary	If day is Sat/Sun - 1, else - 0	Demand pattern
hour/month/week_f	datetime	Indicator for calendar features	Datetime pattern

Features	***Type***	***Characteristics***	***Usage***
lag1, lag2	lag	Log price 1 & 2 hrs prior	Last hours comparison
lag12, lag24	lag	Log price 12 & 24 hrs prior	Midday & daily cycle
lag48	lag	Log price 48 hrs prior	Two-day cycle
lag168	lag	Log price 168 hrs prior (same hour, last week)	Weekly seasonality
aillag1, aillag24	lag	Adjusted for log price, 1 hr & 24 hrs prior	AIL impacts hourly and daily price
rollmean24	derived	Rolling mean of log price over 24 hrs	Ravg30 substitute
rollsd24	derived	Rolling std dev of log price over 24 hrs	Pattern recognition
spikecount24	spike	Count of spikes in the last 24 hrs	Spike clustering
hrssincespike	spike	Hours elapsed since last spike (max 168)	Spike detection

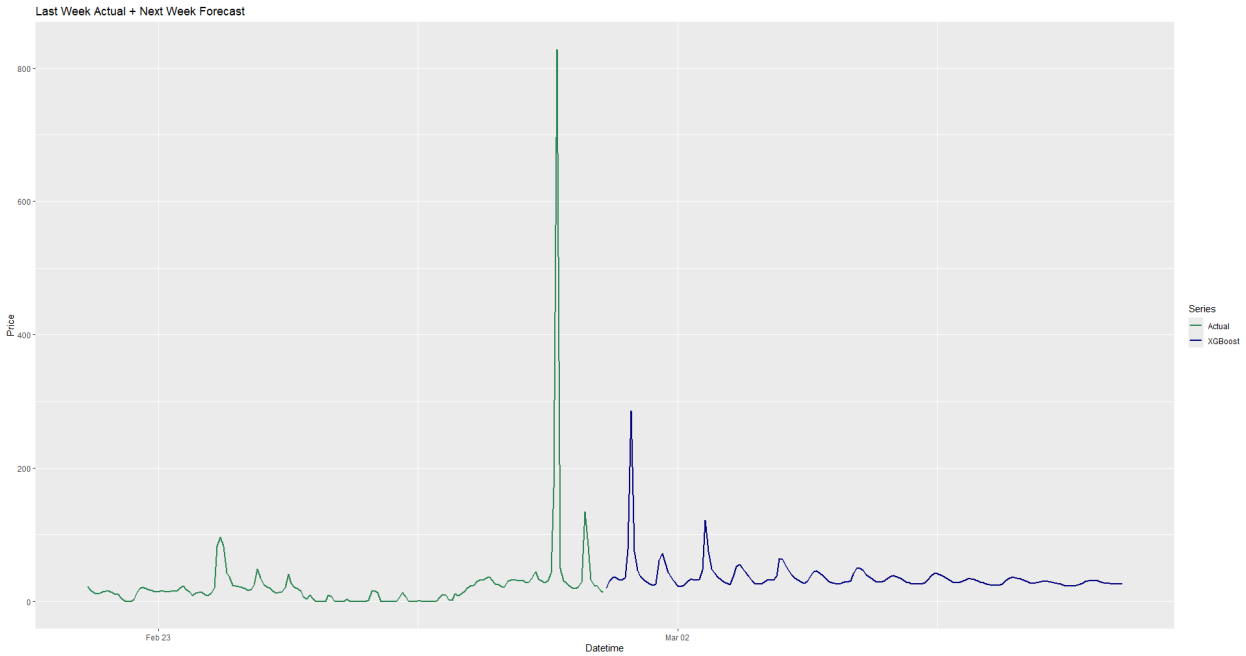
Appendix F: Actual pool prices vs XGBoost test predictions



Appendix G: Residuals of test predictions for XGBoost model



Appendix H: Last week actual prices vs next week future values



Appendix I: How pool price overlaps with metered data